Reminder:

- Reversibility → System can be run in reverse.
- Isothermal → Reversible + constant T
- Adiabatic → Reversible + no flow of heat

Isotherm

$\Delta Q = N k_B T \ln \frac{V_2}{V_1}$

Adiabats

$pV^\gamma = \text{constant}$

The Second Law & Heat engines

Historically people wanted to know how to build machines to do work

* Much much more than this.

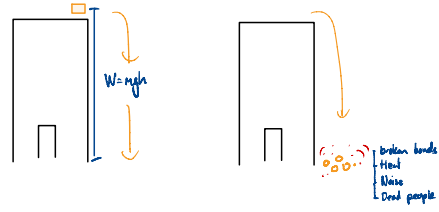
Atkins

The second law is one of the all-time great laws of science, for it illuminates why anything—anything from the cooling of hot matter to the formulation of a thought—happens at all. The

* Heat goes from hot to cold. And in isolation at least, the reverse never happens.
* We are starting to develop a modern statistical understanding for why this.
* Does the historic understanding of this have anything to offer.

Clausius' statement of the second law of thermodynamics
"No process is possible whose sole result is the transfer of heat from a colder to a hotter body."

* Consider a brick ....



work to heat is easy. What about heat to work? Not so much

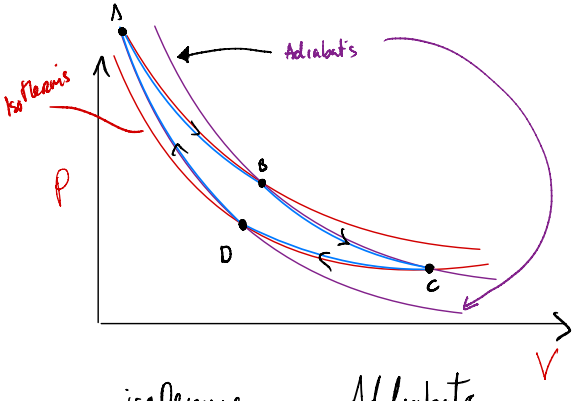
Kelvin's statement of the second law of thermodynamics:
"No process is possible whose sole result is the complete conversion of heat into work."

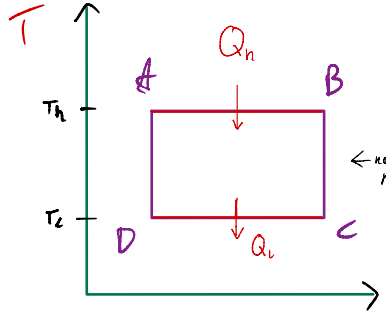
Carnot cycle

Kelvin's statement does not forbid some conversion of heat into work

Define an engine → Cyclic process that converts heat into work

Carnot





$f(pV^\gamma) = \text{constant}$

isotherms	Adiabats
$A \to B$	$B \to C$
$C \to D$	$D \to A$
$\Delta T = 0$	$\Delta Q = 0$

* Operates between T_h & T_c
* Heat only enters during reversible isotherms (or bars)
* S

* $\Delta U = \Delta Q - \Delta W$

Work output $\Rightarrow W = Q_h - Q_c$

Example Find expression for Q_h/Q_L for ideal gas undergoing Carnot cycle in terms of T_h & T_L.

$T_h V_B^{\gamma-1} = T_L V_C^{\gamma-1}$

$A \to B$: $Q_h = RT_h \ln V_B/V_A$

$B \to C$: $\left(\frac{T_h}{T_L}\right) = \left(\frac{V_C}{V_B}\right)^{\gamma-1}$

$C \to D$: $Q_L = -RT_L \ln \frac{V_D}{V_C}$

$D \to A$: $\left(\frac{T_L}{T_h}\right) = \left(\frac{V_A}{V_D}\right)^{\gamma-1}$

$\left(\frac{V_C}{V_B}\right) = \left(\frac{V_D}{V_A}\right) \Rightarrow \frac{V_B}{V_A} = \frac{V_C}{V_D}$

$$\frac{Q_h}{Q_L} = \frac{RT_h \ln V_B/V_A}{-RT_L \ln V_D/V_C} = \frac{T_h}{T_L}$$

$$\frac{Q_h}{Q_L} = \frac{T_h}{T_L}$$

Can use to define temp.

Very important

Concept of efficiency

Ratio of $\frac{\text{what you want to achieve}}{\text{what you have to do to achieve it}}$

For engine $\eta = W/Q_h$

always < 1 since $W < Q_h$

For Carnot the efficiency can be calculated

$$\eta_{Carnot} = \frac{W}{Q_h} = \frac{Q_h - Q_l}{Q_h} = 1 - \frac{Q_l}{Q_h} = 1 - \frac{T_l}{T_h} = \frac{T_h - T_l}{T_h}$$

Powerstations $T_h \sim 800K$ $T_l \sim 300K$

If it was Carnot we could get $1 - \frac{300}{800} \sim \frac{5}{8}$

$\sim 62.\overline{5}\%$

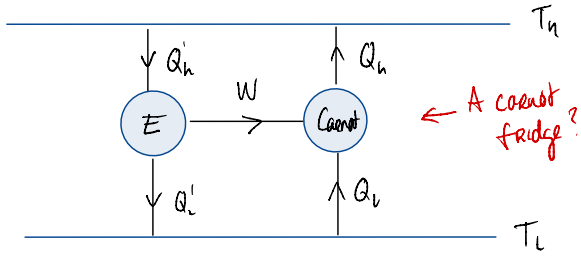
Reality is a bit different.

Carnot's Theorem

Of all the heat engines working between two given temperatures, none is more efficient than a Carnot engine.

$\Rightarrow$ Carnot's theorem is itself a statement of the second Law

Proof: Imagine engine E is an engine more efficient than a Carnot engine. A Carnot Engine is reversible so ... we run it in reverse & connect it up to E as follows



Since $\eta_E > \eta_C$ then $\frac{W}{Q_h'} > \frac{W}{Q_h}$ and so $Q_h > Q_h'$

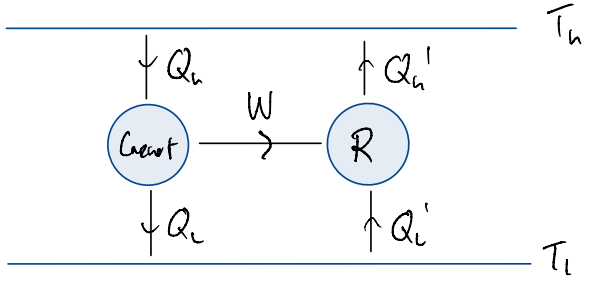
First Law $\Rightarrow$ $W = Q_h' - Q_l' = Q_h - Q_l$ so that

$$Q_h - Q_h' = Q_l - Q_l'$$
$$>0 \qquad >0 \;!!$$

{ Taking Q out of T_l & dumping it in T_h }

So Carnot engine is as efficient as you can get.

Corollary: All reversible engines between two temps. have same efficiency η_{Carnot}



Suppose R is reversible → hook it up to Carnot and run in reverse.

$$\eta_C \geq \eta_R$$

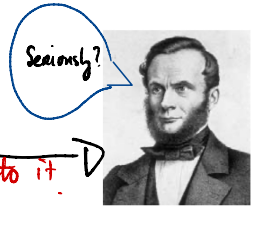
$$\eta_C = \frac{W}{Q_h} \geq \frac{W}{Q_h'} \Rightarrow Q_h' \geq Q_h$$

$$Q_h - Q_h' \leq 0$$

$$W = Q_h - Q_L = Q_h' - Q_L'$$

$$Q_h - Q_h' = Q_L - Q_L'$$

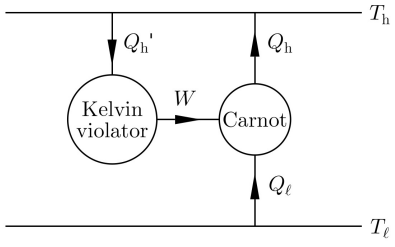
$$\leq 0 \quad \Rightarrow \quad < 0$$



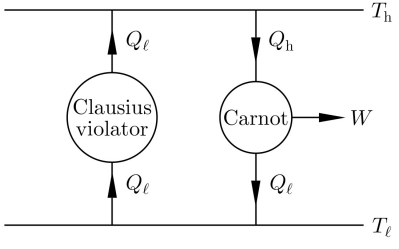
⇒ More heat leaves cold reservoir than goes into it.

All reversible engines have same efficiency $\frac{T_h - T_L}{T_h}$

Equivalence of Clausius & Kelvin statements.



$\Rightarrow$ violates Clausius' statement



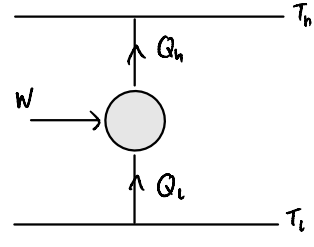
$\Rightarrow$ violates Kelvin's statement

Examples of engines running backwards

(a) A refrigerator:

$$\eta = \frac{Q_L}{W} = \frac{Q_L}{Q_h - Q_L}$$

$$\eta_{\text{Carnot-fridge}} = \frac{T_L}{T_h - T_L}$$ (which can be greater than 1)

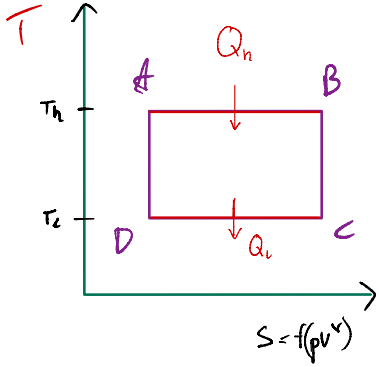


(b) A heat pump:

$$\eta_{\text{pump}} = \frac{Q_h}{W} \Rightarrow \eta_{\text{Carnot-pump}} = \frac{T_h}{T_h - T_L} > 1$$

Clausius Theorem

* Consider a Carnot cycle.



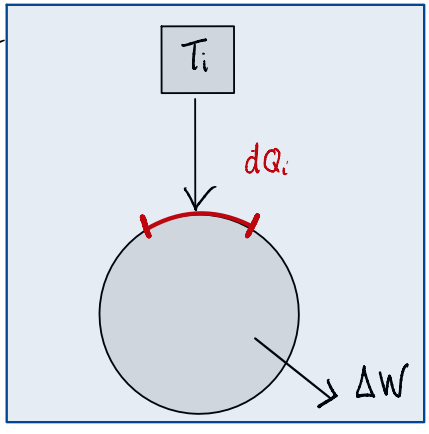
$$\frac{Q_h}{Q_c} = \frac{T_h}{T_c}$$

$$\sum_{cycle} \frac{\Delta Q_{rev}}{T} = \frac{Q_h}{T_h} - \frac{Q_c}{T_c} = 0$$

* Replacing sum by integral we have $\oint \frac{dQ_{rev}}{T} = 0$ for this Carnot cycle.

* Real engines more complicated. Heat enters and exits at various points in a cycle at different temperatures - [sequence of different reservoirs]

*

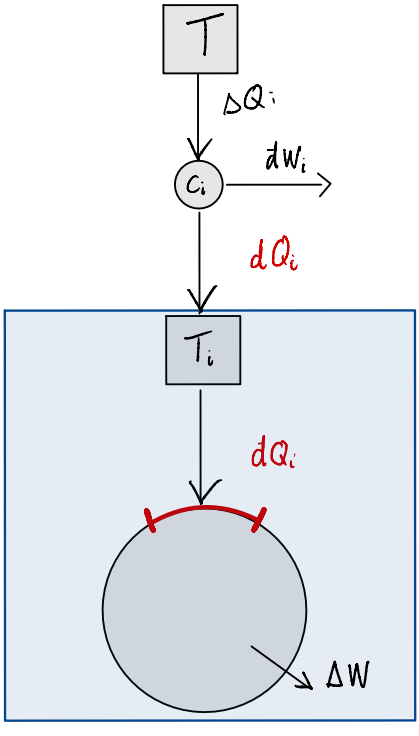


Heat dQ_i enters at a particular point. Over the cycle work extracted is

$\Delta U = \Delta Q - \Delta W$

$$\Delta W = \sum_{cycle} dQ_i \quad (1)$$

"An efficiency"
"A reversibility"



A bunch of Carnot cycles

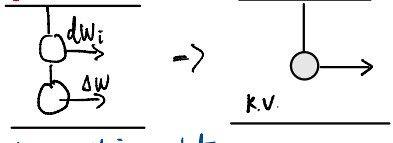
$$\frac{Q_l}{T_l} = \frac{Q_h}{T_h}$$

$$\frac{dQ_i}{T_i} = \frac{\Delta Q_i}{T} = \frac{dQ_i + dW_i}{T}$$

Rearranging

$$dW_i = dQ_i \left(\frac{T}{T_i} - 1\right) \quad (2)$$

Zooming out this looks like



Looks like a Kelvin violation.

* So we demand that $\Delta W + \sum_{cycle} dW_i \leq 0$ (3)

* Using (1), (2) & (3) we have $\sum_{cycle} dQ_i + \sum_{cycle} dQ_i \left(\frac{T}{T_i} - 1\right) \leq 0$

$$\Rightarrow T \sum_{cycle} \frac{dQ_i}{T_i} \leq 0$$

$$\Rightarrow \sum_{cycle} \frac{dQ_i}{T_i} \leq 0 \qquad T > 0$$

$$\Rightarrow \oint \frac{dQ}{T} \leq 0$$

Clausius Theorem

For any closed cycle $\oint \frac{dQ}{T} \leq 0$,

with the equality holding only for reversible cycles.

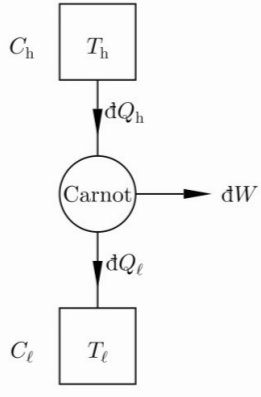


Fig. 13.11 A Carnot engine shown schematically. In diagrams such as this one, the arrows are labelled with the heat and work flowing in one cycle of the engine.

Example 13.5

Two bodies with temperature-independent heat capacities C_h and C_ℓ are used as reservoirs for a Carnot heat engine (see Fig. 13.11). Derive an expression for the total work obtainable.

Solution: In an infinitesimal change we have that

$$\text{đ}Q_\mathrm{h} = -C_\mathrm{h}\,\mathrm{d}T_\mathrm{h} \tag{13.35}$$

$$\text{đ}Q_\ell = C_\ell\,\mathrm{d}T_\ell, \tag{13.36}$$

and for a Carnot engine we have that

$$\frac{\text{đ}Q_\mathrm{h}}{T_\mathrm{h}} = \frac{\text{đ}Q_\ell}{T_\ell}, \tag{13.37}$$

and integrating gives $\int_{T_\ell}^{T_\mathrm{f}} \frac{\text{đ}Q_\ell}{T_\ell} = -\int_{T_\mathrm{h}}^{T_\mathrm{f}} \frac{\text{đ}Q_\mathrm{h}}{T_\mathrm{h}}$ and hence

$$C_\ell \ln\frac{T_\mathrm{f}}{T_\ell} = -C_\mathrm{h}\ln\frac{T_\mathrm{f}}{T_\mathrm{h}}, \tag{13.38}$$

where T_f is the final temperature of each reservoir. Thus

$$T_\mathrm{f}^{C_\mathrm{h}+C_\ell} = T_\mathrm{h}^{C_\mathrm{h}} T_\ell^{C_\ell}. \tag{13.39}$$

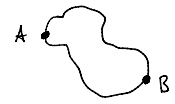
The total heat extracted from each reservoir is $\Delta Q_\mathrm{h} = C_\mathrm{h}(T_\mathrm{h} - T_\mathrm{f})$ and $\Delta Q_\ell = C_\ell(T_\mathrm{f} - T_\ell)$ respectively and so the total work is

$$\Delta W = \Delta Q_\mathrm{h} - \Delta Q_\ell = C_\mathrm{h}T_\mathrm{h} + C_\ell T_\ell - (C_\mathrm{h} + C_\ell)T_\mathrm{f}. \tag{13.40}$$

Entropy!

Clausius says $\oint \frac{dQ_{rev}}{T} = 0$

so $\int_A^B \frac{dQ_{rev}}{T}$ is path independent.



$\frac{dQ_{rev}}{T}$ is an exact differential → new state function called entropy.

We define entropy as $dS = \frac{dQ_{rev}}{T}$

so that

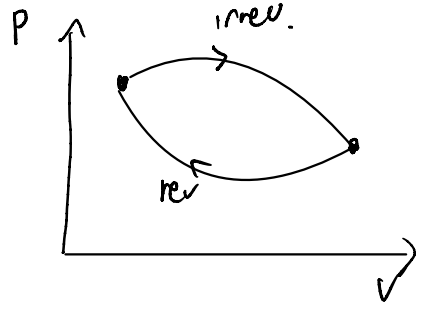
$$S(B) - S(A) = \int_A^B \frac{dQ_{rev}}{T}$$

For adiabatic processes (reversible adiabatic)

$dQ_{rev} = 0 \Rightarrow$ no change in entropy (isentropic)

Irreversible change

$$\oint \frac{dQ_{rev}}{T} = 0$$



Clausius $\oint \frac{dQ}{T} \leq 0$

$$= \int_A^B \frac{đQ}{T} + \int_B^A \frac{đQ_{rev}}{T} \leq 0$$

$$\Rightarrow \int_A^B \frac{dQ}{T} \leq \int_A^B \frac{đQ_{rev}}{T}$$

True for all A & B

In general $dS = \frac{đQ_{rev}}{T} \geq \frac{đQ}{dT}$

Consider a thermally isolated system

... since $đQ = 0$

then $\boxed{dS \geq 0}$ — Very important

Application to the Universe
Assuming that the Universe can be treated as an isolated system, the first two laws of thermodynamics become:

(1) U_{Universe} = constant.
(2) S_{Universe} can only increase.

Example 14.1

A large reservoir at temperature T_R is placed in thermal contact with a small system at temperature T_S. They both end up at the temperature of the reservoir, T_R. The heat transferred from the reservoir to the system is $\Delta Q = C(T_\mathrm{R} - T_\mathrm{S})$, where C is the heat capacity of the system.

- If $T_\mathrm{R} > T_\mathrm{S}$, heat is transferred from reservoir to system, the system warms and its entropy increases; the entropy of the reservoir decreases, because heat flows out of it.
- If $T_\mathrm{R} < T_\mathrm{S}$, heat is transferred from system to reservoir, the system cools and its entropy decreases; the entropy of the reservoir increases, because heat flows into it.

Let us calculate these entropy changes in detail: The entropy change in the reservoir, which has constant temperature T_R, is

$$\Delta S_\mathrm{reservoir} = \int \frac{đQ}{T_\mathrm{R}} = \frac{1}{T_\mathrm{R}} \int đQ = \frac{\Delta Q}{T_\mathrm{R}} = \frac{C(T_\mathrm{S} - T_\mathrm{R})}{T_\mathrm{R}}, \qquad (14.10)$$

while the entropy change in the system is

$$\Delta S_\mathrm{system} = \int \frac{đQ}{T} = \int_{T_\mathrm{S}}^{T_\mathrm{R}} \frac{C\,\mathrm{d}T}{T} = C \ln \frac{T_\mathrm{R}}{T_\mathrm{S}}. \qquad (14.11)$$

Hence, the total entropy change in the Universe is

$$\Delta S_\mathrm{Universe} = \Delta S_\mathrm{system} + \Delta S_\mathrm{reservoir} = C \left[\ln \frac{T_\mathrm{R}}{T_\mathrm{S}} + \frac{T_\mathrm{S}}{T_\mathrm{R}} - 1 \right]. \qquad (14.12)$$

These expressions are plotted in Fig. 14.2 and demonstrate that even though $\Delta S_\mathrm{reservoir}$ and ΔS_system can each be positive or negative, we always have that

$$\Delta S_\mathrm{Universe} \geq 0. \qquad (14.13)$$

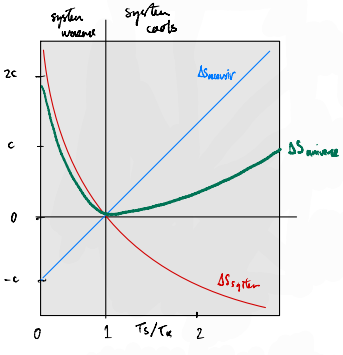


Fig. 14.2 The entropy change in the simple process in which a small system is placed in contact with a large reservoir.

10°C ~ 283K
90°C ~ 363K
50°C ~ 323K

Example 5.2

The ~~contacts~~ contents of a cup containing 0.57kg of cold water at 10°C is poured into another cup containing the same amount of hot water at 90°C. How much entropy is created in the process?

Answer:

1. Since the amount of hot and cold water are the same, the final temperature will be the average of T_{hot} and T_{cold}, ie $T_f = 50°\text{C}$.

2. Let us first assume the two do **not** mix (eg. they are left in separate cups but in thermal contact).

   Now T is not constant, so we cannot use $\Delta S = \frac{Q}{T}$ directly. However, for a *small* heat transfer $dQ = CdT = mcdT$, we can take the denominator as fixed, ie.

   $$dS = \frac{mc\,dT}{T}$$

   Integrating these small changes, we find

   $$\Delta S_{\text{hot}} = \int_{T_{\text{hot}}}^{T_{\text{f}}} \frac{mcdT}{T} = mc \ln \frac{T_{\text{f}}}{T_{\text{hot}}} = (0.57\text{kg})(4190\text{J/Kg}) \ln \frac{323\text{K}}{363\text{K}} = -279\text{J/K}$$

   $$\Delta S_{\text{cold}} = \int_{T_{\text{cold}}}^{T_{\text{f}}} \frac{mcdT}{T} = mc \ln \frac{T_{\text{f}}}{T_{\text{cold}}} = (0.57\text{kg})(4190\text{J/Kg}) \ln \frac{323\text{K}}{283\text{K}} = +316\text{J/K}$$

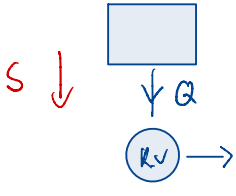
   The total change in entropy is $\Delta S = \Delta S_{\text{hot}} + \Delta S_{\text{cold}} \underline{+37\text{J/K}}$

3. What happens when two bodies of water mix?

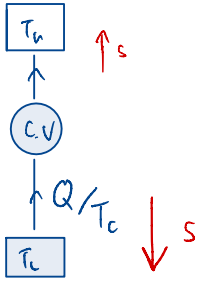
   For an ideal gas consisting of identical atoms we will see later that there is *no* change in entropy when two quantities of the *same* gas mix. This is because the atoms of one body are already indistinguishable from those of the other, so no additional ways of arranging them are brought into being when they are allowed to mix. These considerations hold for *any* gas or liquid, so $\underline{\Delta S_{\text{mix}} = 0}$

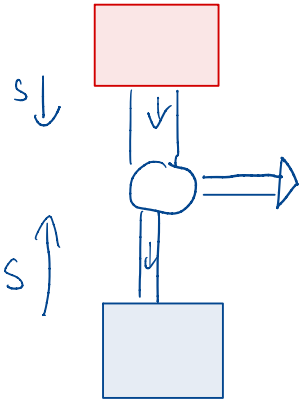
Carnot — The circle of
Clausius — Kelvin — Thermodynamics
Entropy — truth

Kelvin



Clausius





Suppose Q_h leaves the hot source as heat: the entropy falls by Q_h/T_h. Suppose Q_c is discarded into the cold sink. The entropy increases by Q_c/T_c. For the overall change in entropy to be positive, the minimum amount of heat to discard is such that $Q_c/T_c = Q_h/T_h$ or therefore $Q_c = Q_h T_c/T_h$. This means that maximum work that can be done is $Q_h - Q_c$ or $Q_h(1 - T_c/T_h)$.

The efficiency is this work divided by the heat supplied Q_h $\Rightarrow \eta = 1 - T_c/T_h$. "Carnot!"

Reminder

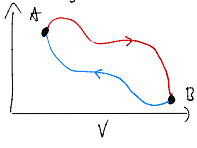
①

Kelvin — Clausius (the same)

Carnot cycle → $\frac{Q_h}{T_h} = \frac{Q_c}{T_c} \Rightarrow \sum_{\text{cycle}} \frac{Q_{rev}}{T} = 0$

$\Rightarrow \sum_{\text{rev. cycle}} \frac{Q_{rev}}{T} = 0$

Clausius Theorem $\oint \frac{dQ}{T} \leq 0$ with $\oint \frac{dQ_{rev}}{T} = 0$



②

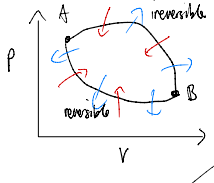
$\int \frac{dQ_{rev}}{T} = 0 \Rightarrow \int_A^B \frac{dQ_{rev}}{T} + \int_B^A \frac{dQ_{rev}}{T} = 0$

$\int_A^B \frac{dQ_{rev}}{T} = \int_A^B \frac{dQ_{rev}}{T}$

Change in quantity $\frac{dQ_{rev}}{T}$ independent of path

So we define $dS = \frac{dQ_{rev}}{T}$

③



$\rightarrow \oint \frac{dQ}{T} \leq 0$

$\int_A^B \frac{dQ}{T} + \int_B^A \frac{dQ_{rev}}{T} \leq 0$

$\int_A^B \frac{dQ}{T} \leq \int_A^B \frac{dQ_{rev}}{T}$ or $\int_A^B dS$

④

$\Rightarrow \int_A^B dS \geqslant \int_A^B \frac{dQ}{T}$

Consider a closed (thermally isolated) system

... since $dQ = 0$

then, because true for any paths & any A & B

$dS \geqslant 0$ — Very important

Application to the Universe
Assuming that the Universe can be treated as an isolated system, the first two laws of thermodynamics become:

(1) U_{Universe} = constant.
(2) S_{Universe} can only increase.

First Law revisited

$$dU = dQ + dW$$

$$dQ_{rev} = T\,dS \quad \Big\} \text{ Entropy definition}$$

$$dW = -p\,dV$$

$$\boxed{dU = T\,dS - p\,dV} \quad \text{Super important}$$

* Where it is worth stressing that dQ is reversible

* However since all functions in ⊛ are functions of state / variables of state → it also holds for irreversible process also !!!

Irreversible change we have $\frac{dQ}{T} \leq dS \Rightarrow T\,dS \geq dQ$, but this is compensated by $dW \geq -p\,dV$. We always have

$$dU = T\,dS - p\,dV$$

$T\,dS$: Intensive ↓ T, S are a bit like forces

$p\,dV$: extensive ↓ S, V are a bit like coordinates

Energy

$dW = F \cdot dx$

Reminder

Summary		
$dU = đQ + đW$	always true
$đQ = T\,dS$	only true for reversible changes
$đW = -p\,dV$	only true for reversible changes
$dU = T\,dS - p\,dV$	always true
For irreversible changes:	$đQ \leq T\,dS$,	$đW \geq -p\,dV$

Heat absorbed by the system is less for irreversible change
Heat emitted by the system is greater for irreversible change

Work done on the system greater for irreversible change
(Work done by the system is less for irreversible change)

Most efficient engine is a reversible one

Mathematically : $dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$

Identify $T = \left(\frac{\partial U}{\partial S}\right)_V \qquad P = -\left(\frac{\partial U}{\partial V}\right)_S$

Aside

Ratio : $\frac{P}{T} = -\left(\frac{\partial U}{\partial V}\right)_S \Big/ \left(\frac{\partial U}{\partial S}\right)_V = -\left(\frac{\partial U}{\partial V}\right)_S \left(\frac{\partial S}{\partial U}\right)_V$ (Reciprocal Theorem)

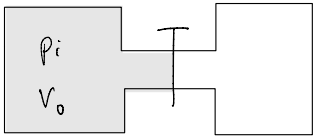
$= \left(\frac{\partial S}{\partial V}\right)_U$ (reciprocity theorem) $\Rightarrow \frac{1}{T} = \left(\frac{\partial S}{\partial U}\right)_V$

* Problem set 3

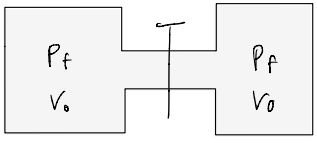
$\left(\frac{\partial x}{\partial z}\right)_y = \frac{1}{\left(\frac{\partial z}{\partial x}\right)_y}$ | $\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1$

together : $\left(\frac{\partial x}{\partial y}\right)_z = -\left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x$

The Joule expansion



1 mole of ideal gas



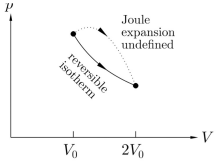


Fig. 14.5 The Joule expansion between volume V_0 and volume $2V_0$ and a reversible isothermal expansion of a gas between the same volumes. The path in the p–V plane for the Joule expansion is undefined, whereas it is well defined for the reversible isothermal expansion. In each case however, the start and end points are well defined. Since entropy is a function of state, the change in entropy for the two processes is the same, regardless of route.

Ideal gas $\quad p_i V_0 = R T_i$

$p_f 2V_0 = R T_f$

Thermally isolated $\quad \Delta U = 0 \Rightarrow \Delta T = 0$

$p_i V_0 = 2 p_f V_0$

$p_i = 2 p_f$

$U = \frac{3}{2} R T \qquad pV = n_M R T$

Isothermal $\Delta T = 0$

Adiabatic $\Delta Q = 0$

Just after partition removed → pressure, Vol undefined.

Isothermal $\quad \Delta T = \Delta U = 0$ (ideal gas)

$T dS = p dV$

$$\Delta S = \int_i^f dS = \int_{V_0}^{2V_0} \frac{p\,dV}{T} = \int_{V_0}^{2V_0} \frac{R\,dV}{V} = R \ln\left(\frac{2V_0}{V_0}\right) = R \ln 2$$

Ex 14.3

What is the change of entropy in the gas, surroundings & the universe.

ΔS_{gas} → same for Joule expansion, reversible isotherm

* Reversible → implies no increase of entropy in the universe →

$\Delta S_{gas} = R \ln 2$

$\Delta S_{surround} = -R \ln 2$

$\Delta S_{universe} = \Delta S_{gas} + \Delta S_{surround} = 0$

* Entropy of surroundings goes down!
  ↳ it is not isolated connected to the system

* For Joule expansion, ⇒ system is thermally isolated

$\Delta S_{gas} = R \ln 2$

$\Delta S_{surround} = 0$

$\Delta S_{universe} = R \ln 2$

A paradox?

- In the Joule expansion, the system is thermally isolated so no heat can be exchanged: $\Delta Q = 0$.
- No work is done: $\Delta W = 0$.
- Hence $\Delta U = 0$ (so for an ideal gas, $\Delta T = 0$).
- But if $\Delta Q = 0$, doesn't that imply that $\Delta S = \Delta Q/T = 0$?

The above reasoning is correct, until the very end: the answer to the question in the last point is NO! The equation $\mathrm{d}Q = T\mathrm{d}S$ is only true for reversible changes. In general $\mathrm{d}Q \leq T\mathrm{d}S$, and here we have $\Delta Q = 0$ and $\Delta S = R \ln 2$, so we have that $\Delta Q \leq T\Delta S$.

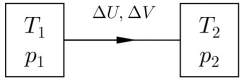


Fig. 14.3 Two systems, 1 and 2, which are able to exchange volume and internal energy.

$$dU = T\,dS - p\,dV$$

$$dS = \frac{1}{T}dU + \frac{p}{T}dV$$

$$dS = \left(\frac{\partial S}{\partial U}\right)_V dU + \left(\frac{\partial S}{\partial V}\right)_U dV$$

Example 14.2

Consider two systems, with pressures p_1 and p_2 and temperatures T_1 and T_2. If internal energy ΔU is transferred from system 1 to system 2, and volume ΔV is transferred from system 1 to system 2 (see Fig. 14.3), find the change of entropy. Show that equilibrium results when $T_1 = T_2$ and $p_1 = p_2$.

Solution:

Equation 14.18 can be rewritten as

$$\mathrm{d}S = \frac{1}{T}\mathrm{d}U + \frac{p}{T}\mathrm{d}V. \tag{14.24}$$

If we now apply this to our problem, the change in entropy is then straightforwardly

$$\Delta S = \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\Delta U + \left(\frac{p_2}{T_2} - \frac{p_1}{T_1}\right)\Delta V. \tag{14.25}$$

Equation 14.9 shows that the entropy always increases in any physical process. Thus, when equilibrium is achieved, the entropy will have achieved a maximum, so that $\Delta S = 0$. This means that the joint system cannot increase its entropy by further exchanging volume or internal energy between system 1 and system 2. $\Delta S = 0$ can only be achieved when $T_1 = T_2$ and $p_1 = p_2$.

https://www.youtube.com/watch?v=tuxbMfKO9Pg

in this house we obey the laws of thermodynamics!

A statistical definition of entropy

$$dS = dQ_{rev}/T$$

Recall however that we defined in terms of the multiplicities Ω.

$$\frac{1}{T} = k_B \frac{d \ln \Omega}{dE}$$

* We also have defined above $\frac{1}{T} = \left(\frac{dS}{dU}\right)_V$

* This motivates the identification

$$S = k_B \ln \Omega$$

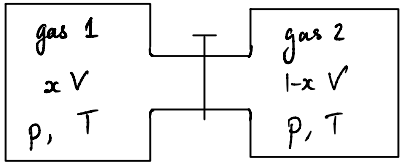
Entropy of a system in a particular macrostate in terms of Ω, the number of microstates associated with this macrostate. Macrostate of fixed energy - micro canonical ensemble

Example Joule expansion revisited

- Each molecule can be in either left or right side
- For one mole there are 2^{N_A} ways of placing them
- For case when gas container is twice as big that means $\Omega_{2V_0} = \Omega_{V_0} 2^{N_A}$ so additional entropy

$$\Delta S = k_B \ln \Omega_{V_0} 2^{N_A} - k_B \ln \Omega_{V_0}$$
$$= k_B \ln 2^{N_A} = k_B N_A \ln 2 = R \ln 2$$

Entropy of Mixing



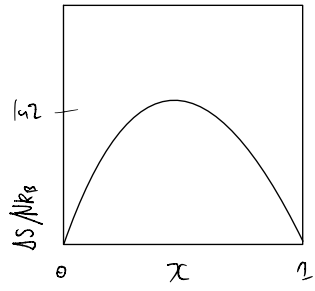
- 2 ideal gases
- $p = \left(\frac{N}{V}\right) k_B T$
- # molecules of gas 1 = xN
  # " " " 2 = $(1-x)N$

- Like in Joule expansion we imagine mixing reversibly
- Energy doesn't change $U = \frac{3}{2} N k_B T$ (isothermal $\Delta T = 0$)
- First Law $TdS = pdV \Rightarrow dS = \left(\frac{p}{T}\right) dV = N k_B \frac{dV}{V}$

$$\Delta S = xN k_B \int_{xV}^{V} \frac{dV_1}{V_1} + (1-x) N k_B \int_{(1-x)V}^{V} \frac{dV_2}{V_2}$$

* And so

$$\Delta S = -N k_B \left(x \ln x + (1-x) \ln (1-x) \right)$$



No entropy increase if $x = 0$ or $x = 1$ — equilibrium state.

Max entropy when $x = \frac{1}{2}$ ⇒ Maximum uncertainty about position of a particle

* After opening the top → each molecule can exist in additional microstates → for every 1 on left there is another on the right.

$$\Omega_M \to 2^N \Omega_{n.m}$$

$$\Delta S = k_B \ln \left(2^{Nx} \frac{\Omega_f'}{\Omega_i'} \right) + k_B \ln \left(2^{N(1-x)} \frac{\Omega_f''}{\Omega_i''} \right)$$

$$= k_B N \frac{1}{2} \ln 2 + k_B \frac{1}{2} N \ln 2$$

$$= k_B N \ln 2.$$

Q. Suppose both gases were the same? What would be the change in entropy?